

34870 Lab B – Microphone scattering – E26

Objectives:

- a) *Introduce measurements in the anechoic chamber,*
- b) *Observe the effect of a microphone-shaped body when exposed to a sound wave in a free field environment,*
- c) *Compare the result with a numerical simulation and with actual microphone free-field correction curves from datasheets.*

Equipment:

- Desktop PC with internal sound card,
- UMIK MiniDSP microphone, USB cable and stand,
- Audio amplifier,
- Scaled-up microphone mock-up made in wood with stand,
- Loudspeaker with cabling,
- Measuring tape, ruler with angle markings / printed protractor.

Warnings:

- *Read and understand this exercise description before you start.*
- ***The absorbing wedges in the anechoic chamber are very fragile and expensive. Pay extra attention not to touch and damage them, especially when you move equipment in and out of the room. Be careful with gaps in the wire mesh.***
- ***Walk slowly inside the chamber. The microphone mock-up is heavy and may move or fall off. Move it carefully, preferably by two people.***
- *Be careful not to drop screws, nuts or small objects through the net in the chamber when setting up the loudspeaker stand and screens. **They cannot be recovered.***
- *Loudspeaker: The moving parts of the driver units are fragile, especially the tweeter which can be damaged even by a light touch. **Please do not touch the drivers.***
- *Cables: There will be long USB and loudspeaker cables laid from the anechoic room. Do not twist the cables or step on them. Do not close the chamber door completely to allow cables to pass through. Before closing the control room door, you must fold the loudspeaker cable inside and leave it on the table outside to avoid damage by the door.*

Evaluation of the exercise

The exercise work is made in groups of ideally three students. However, the evaluation is individual using a quiz in DTU Learn. The responses do not need to be the same for all students in the group. The quiz covers Lab B and Lab C. Submit the quiz as soon as possible and **no later than the deadline Monday 5th October.**

Part 0: Anechoic chamber introduction

The measurements in this exercise are made inside an *anechoic chamber*. This facility has two properties: i) good insulation from the surroundings, and ii) very high absorption of sound waves by the wedges on the walls. The sound field inside the chamber is called *free field*, which means that small sound sources produce nearly spherical waves and the effect of the room is negligible. It is important to note that these properties only hold down to the cutoff frequency of the chamber, 125 Hz in this case, which limits very low frequency measurement of e.g. loudspeakers.

It is possible to test the free field condition inside the chamber. You must remove the microphone mock-up and leave the loudspeaker and UMIK microphone. The sound field produced by the loudspeaker should be close to a spherical wave. You may include one or two more measurements to the procedures in the next section changing the microphone distance to the loudspeaker. The sound pressure magnitude should decrease as $1/r$, following theory.

Part 1: Measurement of the scattering by a cylinder (mic. mock-up)

Introduction

In order to observe how the presence of a microphone changes the sound field, a scaled-up mock-up of a microphone will be used. The mock-up is made of wood. Its surface can be considered acoustically hard. The upscaling allows the use of a regular microphone for sensing the sound pressure close to the centre of the front, representing the diaphragm. The results can be scaled down to the size of an actual microphone by applying the corresponding scale factor (i.e. the ratio of sizes).

Measurement setup

You will measure the effect of the angle of incidence of sound waves on the frequency response of the microphone. We assume that, at the distance used, the spherical waves are nearly plane. The ratio (or difference in dB) of the responses with and without the mock-up must be calculated. You must therefore also measure the response of the setup (loudspeaker + microphone) without the mock-up in the chamber. It is very important that **the positions of microphone and loudspeaker are not changed** throughout the whole measurement, or else the compensation would be incorrect. If you want to make free field test described in Part 0, you must do it independently, before starting the mock-up test.

In order to expose the mock-up to different angles, it must be placed so that the measurement microphone **almost touches the centre of its front**. The microphone should be pointing upwards so that there is room to move the mock-up to angles from 0 to 90 degrees to the incidence direction. This microphone will pick-up the “pressure excitation” of the mock-up. Make some trials before starting the process. Use the measuring tape to measure the dimensions of the mock-up. Measure the frequency response with at least 0, 45, and 90 degree angles of the mock-up. The setup is illustrated in Figure 1.

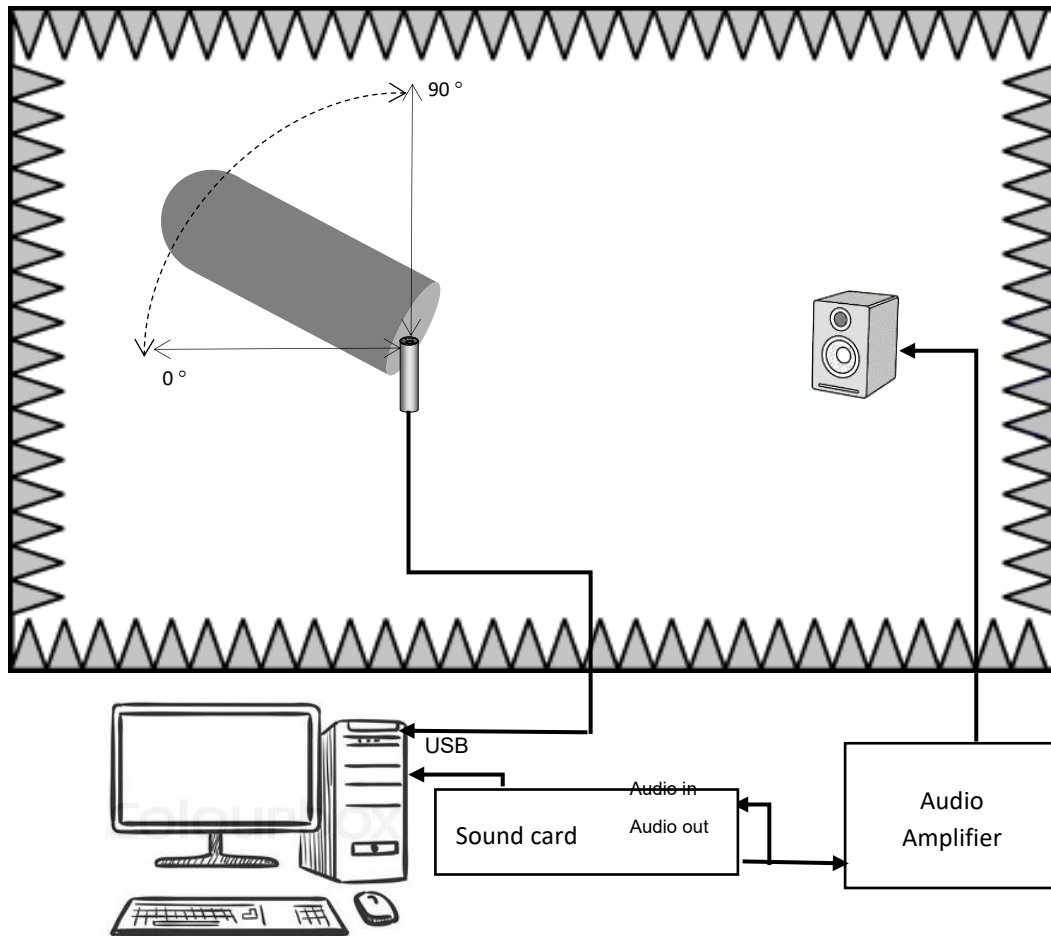


Figure 1- Measurement setup in the anechoic chamber. The mock-up microphone is moved around the measurement microphone.

Measurement procedure and routines

The exercise uses the internal sound card of the desktop PC in the control room. The “line-out” jack has been fed back to the “line-in” jack and sent to the audio amplifier input. The UMIK microphone should be properly placed in the anechoic chamber (see Fig. 1) and connected with the extended USB cable to a USB port in the desktop. Do this before opening Matlab so that the device is registered. You will need to run some trial measurements to find the proper volume levels of the amplifier. Adjust the volume control in the audio amplifier (starting very low) until the amplitude of the measured responses is well over noise. Use the plots generated by the measurement function for these verifications.

You will use the Matlab measurement function `meas_mag_spec2_SoundCard_LabB`. The necessary files are available at DTU Learn (Lab B Module) and are also copied into the lab



computer. When you have all the data, you may collect the results in a memory stick (the computer has no internet connection) and delete the files from the computer.

Matlab command:

```
[fn,specn,f,spec,ch]=  
meas_mag_spec2_SoundCard_LabB(f1,f2,n_oct,fres,Nav,UmikSN,IncAngle)
```

Use the following input parameters:

```
f1=50 Hz      (low frequency limit)  
f2=10000 Hz   (high frequency limit)  
n_oct=24      (number of frequencies per octave)  
fres=1 Hz     (Spectrum frequency resolution)  
Nav=32        (Long averaging to suppress noise)  
UmikSN        (Umik serial number as a string. e.g. '708-0335')  
IncAngle=90   (Angle of incidence on the measurement microphone, NOT MOCKUP ANGLE)
```

The outputs are:

```
fn :      signal frequency vector  
specn:    spectrum values for fn  
f:        full frequency vector  
spec:     full resolution spectrum  
ch:       average time signals
```

As mentioned, the function produces figures showing signals, for control purposes. You must create your own figures with the measurement output: *specn* contains the spectrum at the signal frequencies only, *spec* contains the full spectrum and can reveal the background noise level and any distortion in the measurement. The frequency response $V_{mic}/V_{loudspeaker}$ can be calculated by dividing the two columns of *specn* elementwise (dot).

- For each measurement from 0 to 90 degrees (see previous section) calculate the transfer function between the microphone output and the loudspeaker signal:
 $H_{ang} = specn(:,2) ./ specn(:,1)$
- Do the same for the no mock-up case and get H_{no_mockup} . Divide every angle response H_{ang} by H_{no_mockup} to normalize them: $H_{ang_norm} = H_{ang} ./ H_{no_mockup}$.

Part 2: Scaling of the results and comparison with simulations and datasheets

This part of the exercise can be done after the laboratory work. You must however make sure your measurements are correct before leaving the laboratory.

Simulations

A BEM model of the measurement setup is provided through DTU Learn, for comparison. The model reproduces the experiment assuming a plane wave. The package with the necessary files can be downloaded from DTU Learn. It contains the Matlab function *CylinderPlaneWave*.

You may call it using:

```
[fr,p_centre,p_FP,p_avg]=CylinderPlaneWave(AngleS,CylEnd,see,[r_FP z_FP]);
```

Input parameters:

AngleS: Incidence angle, in radians.
CylEnd: Geometry of the back of the cylinder: 'Round' or 'Flat'
see: If 'yes', three plots are produced: i) the BEM mesh at the lowest frequency, ii) the BEM mesh at the highest frequency, and iii) the pressure at the centre of the face as a function of frequency.
[r_FP z_FP]: (optional) position of a field point.

Output variables:

fr: Vector of frequencies (fixed to 50-5382 Hz in 1/12 octaves)
p_centre: Sound pressure on the centre of the cylinder face for all frequencies.
p_FP: Sound pressure at a given point for all frequencies. Default: 3 cm from centre, change with *[r_FP z_FP]*.
p_avg: Averaged sound pressure over the diaphragm using a parabolic weight, for all frequencies.

Besides the pressure at the centre of the face, other results are produced. Consult the function's help. The setting *CylEnd* also allows the study of the effect of a flat microphone back end. Use the "field point" and "pressure average" calculations to explain possible deviations and get more insight on your results and how the microphone picks up sound.

Scaling

This technique is widely used in linear acoustics, for example by creating small-scale concert hall models where all dimensions are reduced by a factor (*scaled-down*). The corresponding scaled wavelength is also reduced in size, meaning that the frequency is increased in the model by the scale factor.

In Lab B, the microphone mock-up is a *scaled-up* version of a microphone. Obtain the scale factor by dividing the diameter of the mock-up by the diameter of the microphone it represents. The frequency axis for the mock-up results is shifted down by the same factor.

Datasheets

In the following pages of this text there is an excerpt from the Brüel & Kjær Microphone Handbook. It contains the free-field corrections for microphones of sizes 1", ½", ¼" and 1/8" diameters. Obtain the scale factors. Compare the results from the measurement and the simulation with the provided plots from Brüel & Kjær microphones.

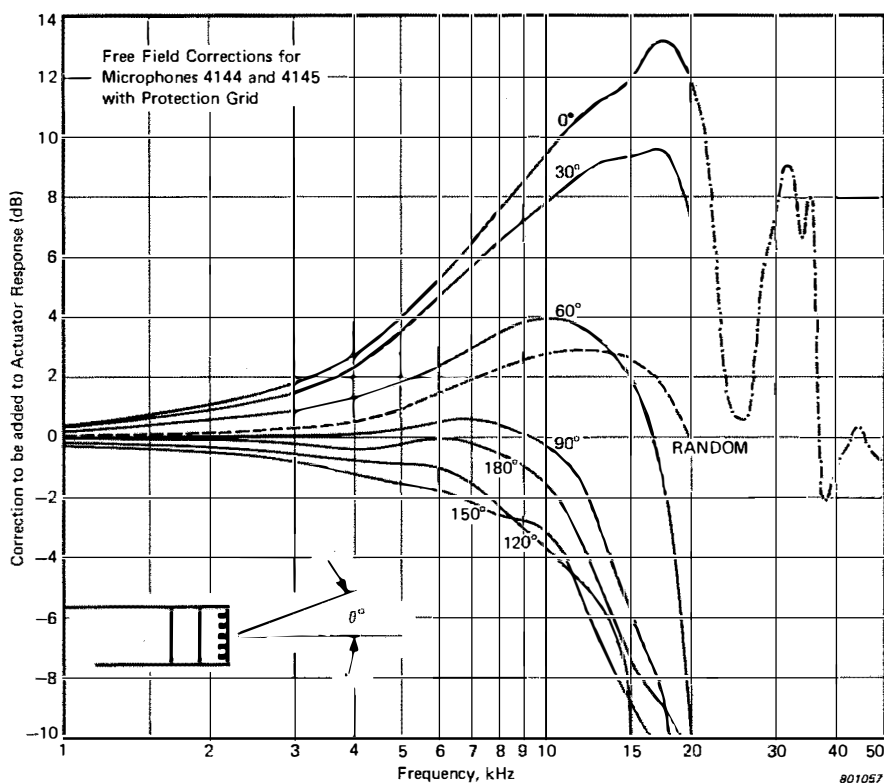


Fig.6.2. Free-field correction curves for one-inch Condenser Microphones Types 4144 and 4145 fitted with normal protection grid

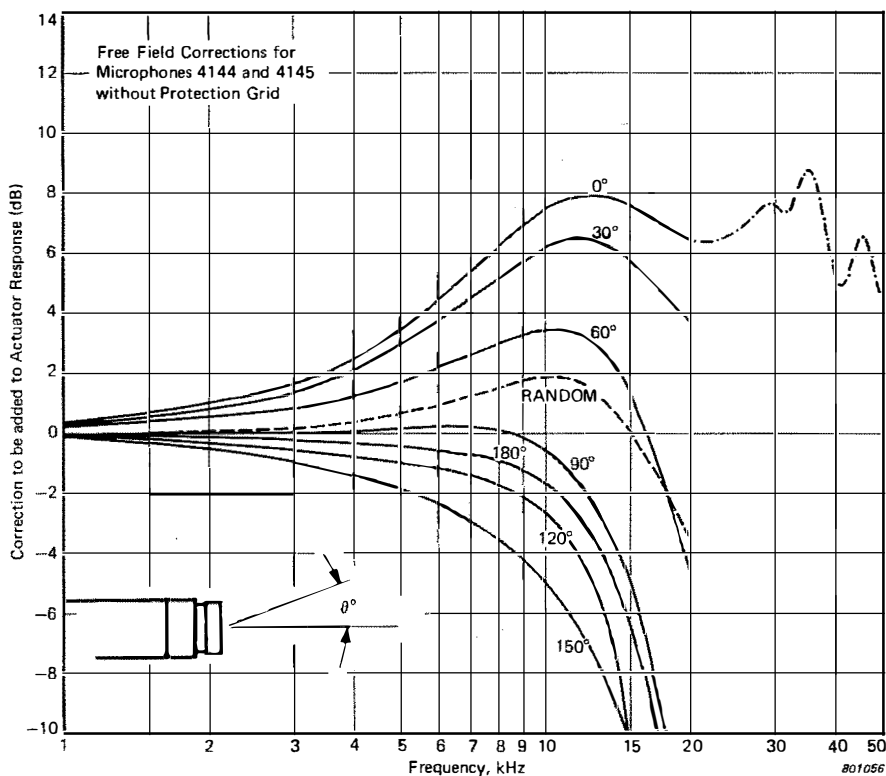


Fig.6.3. Free-field correction curves for one-inch Condenser Microphones Types 4144 and 4145 without normal protection grid

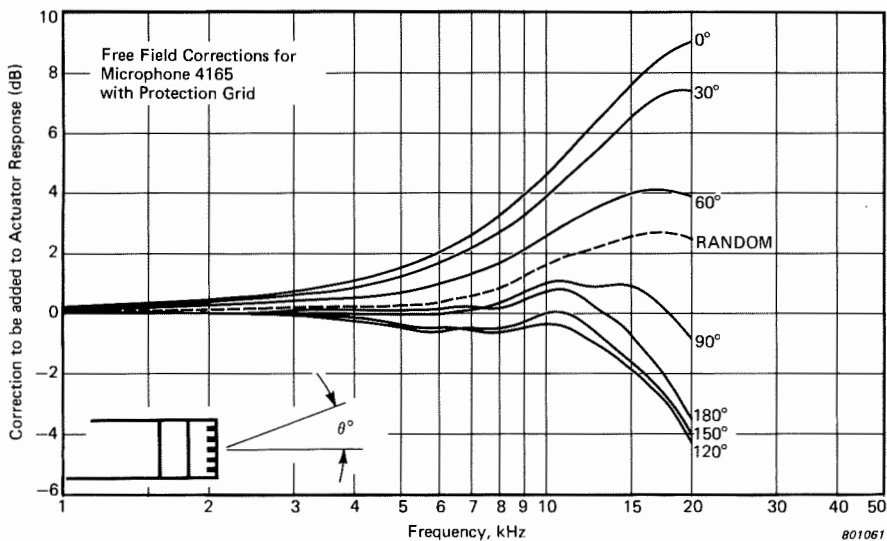


Fig.6.4. Free-field correction curves for half-inch Condenser Microphone Type 4165 fitted with normal protection grid

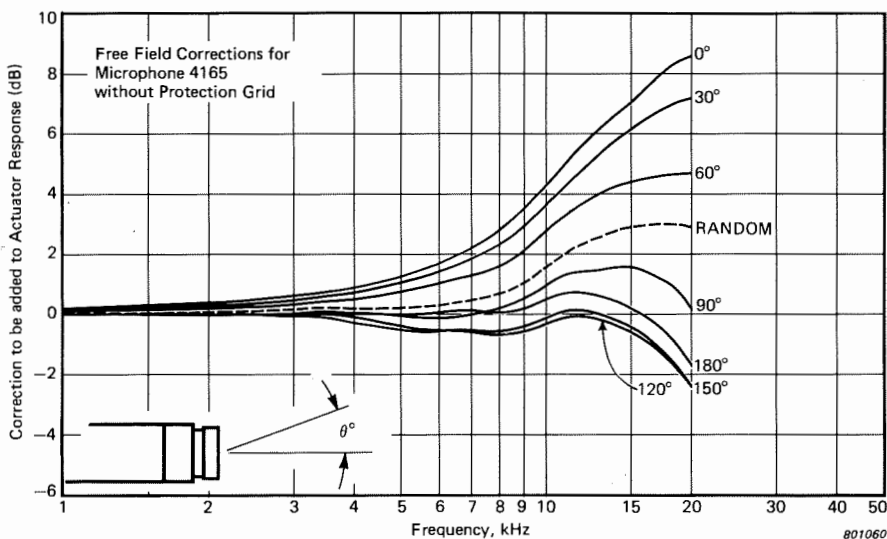


Fig.6.5. Free-field correction curves for half-inch Condenser Microphone Type 4165 without normal protection grid

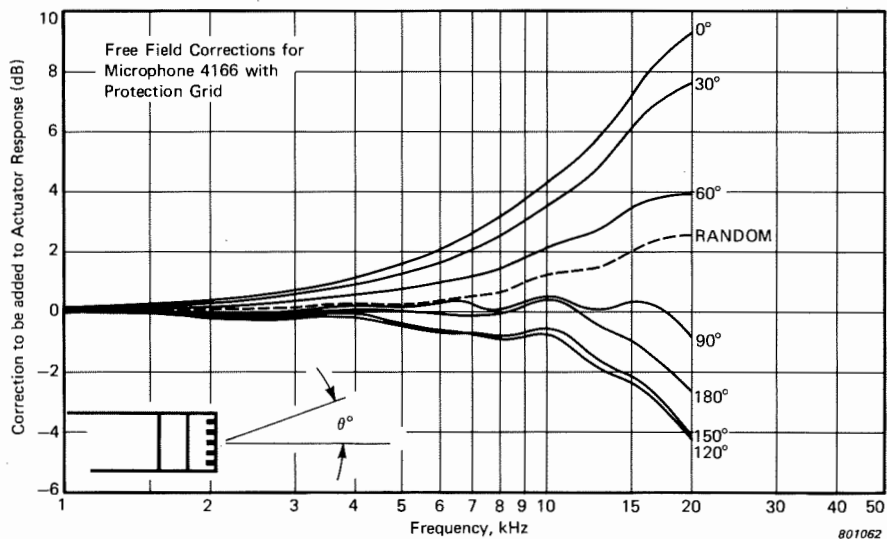


Fig.6.6. Free-field correction curves for half-inch Condenser Microphone Type 4166 fitted with normal protection grid

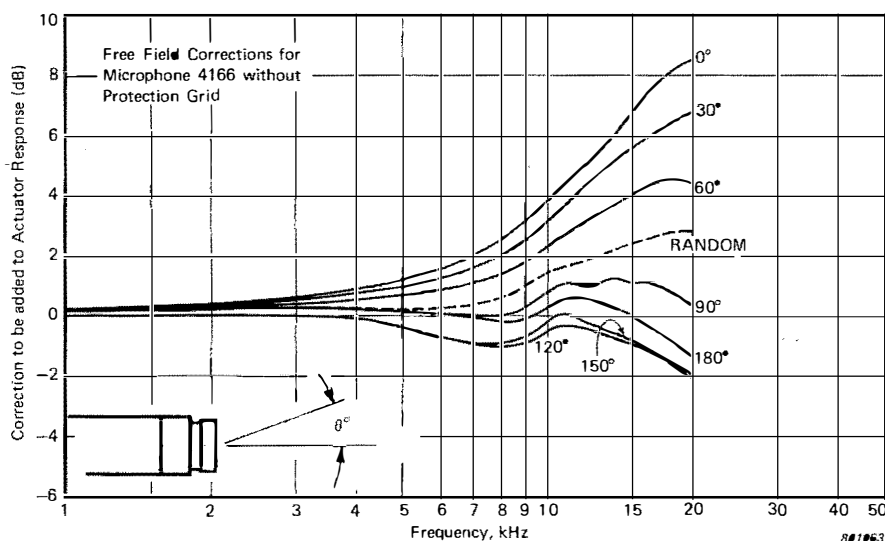


Fig.6.7. Free-field correction curves for half-inch Condenser Microphone Type 4166 without normal protection grid

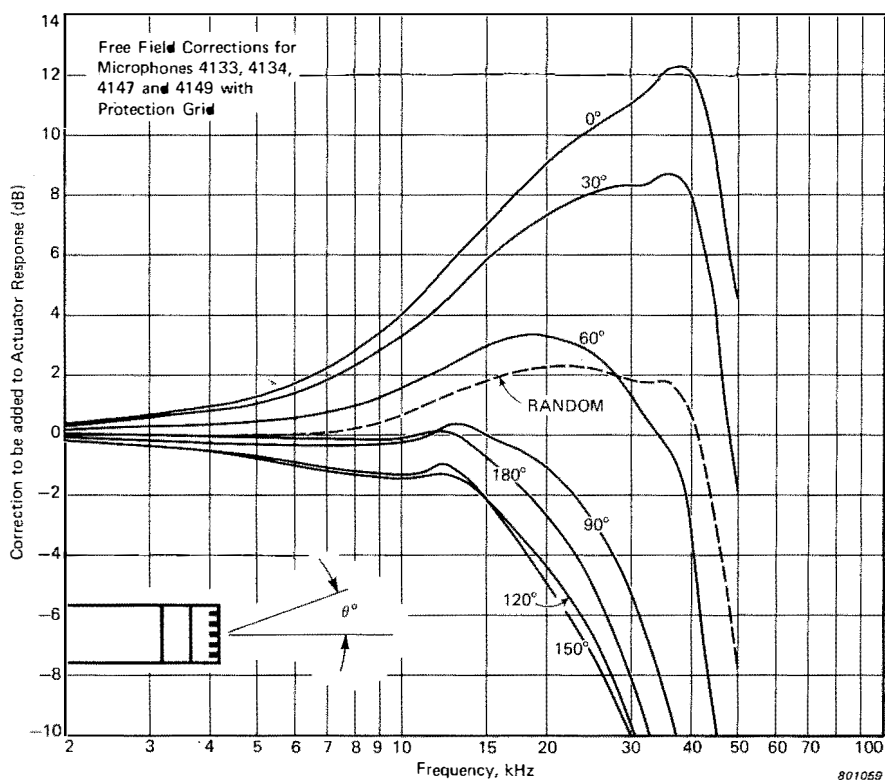


Fig.6.8. Free-field correction curves for half-inch Condenser Microphones Types 4133, 4134, 4147 and 4149 fitted with normal protection grid

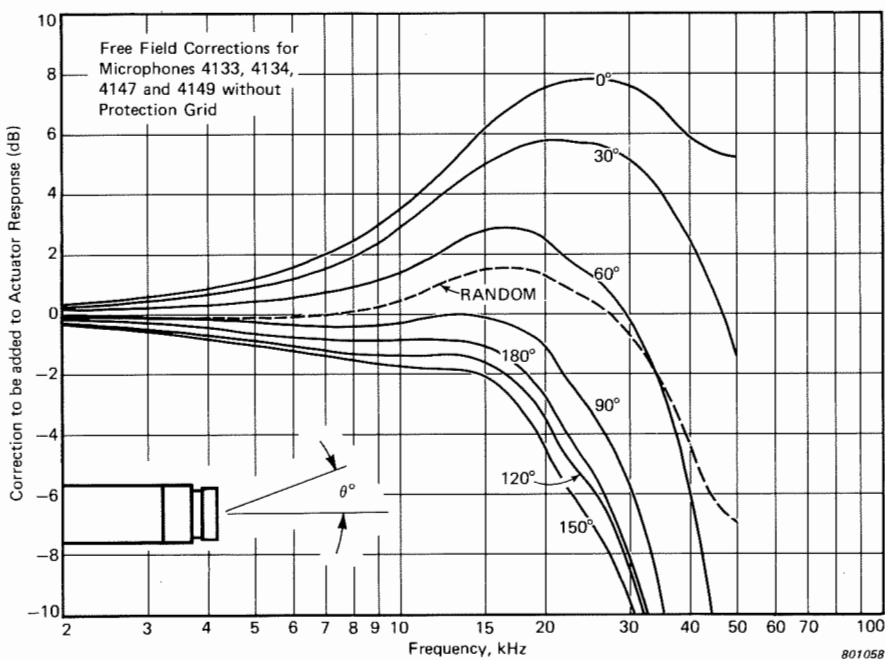


Fig. 6.9. Free-field correction curves for half-inch Condenser Microphones Types 4133, 4134, 4147, 4149 without protection grid

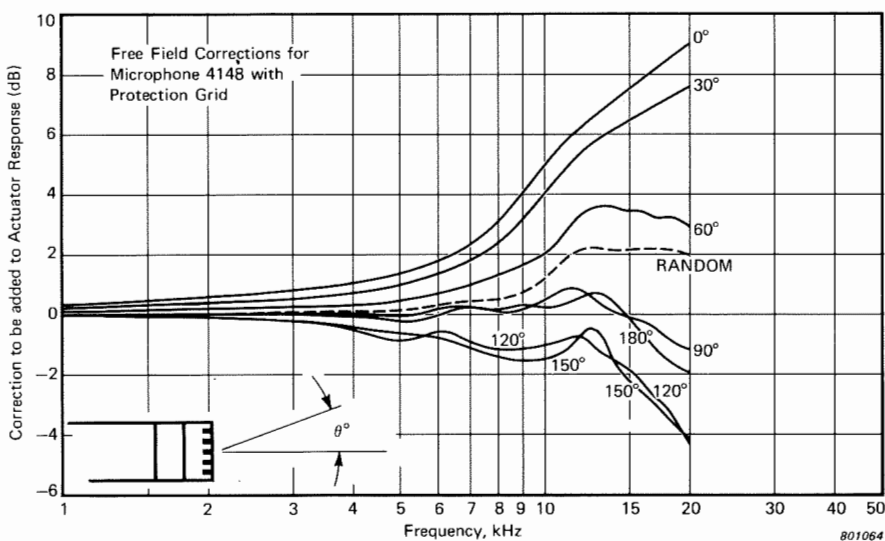


Fig. 6.10. Free-field correction curves for half-inch Condenser Microphone Type 4148 fitted with normal protection grid

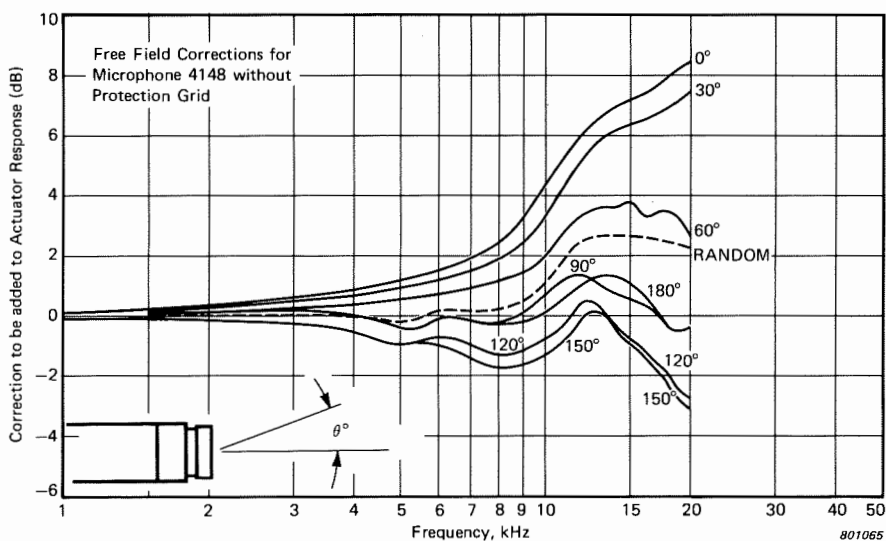


Fig.6.11. Free-field correction curves for half-inch Condenser Microphone Type 4148 without normal protection grid

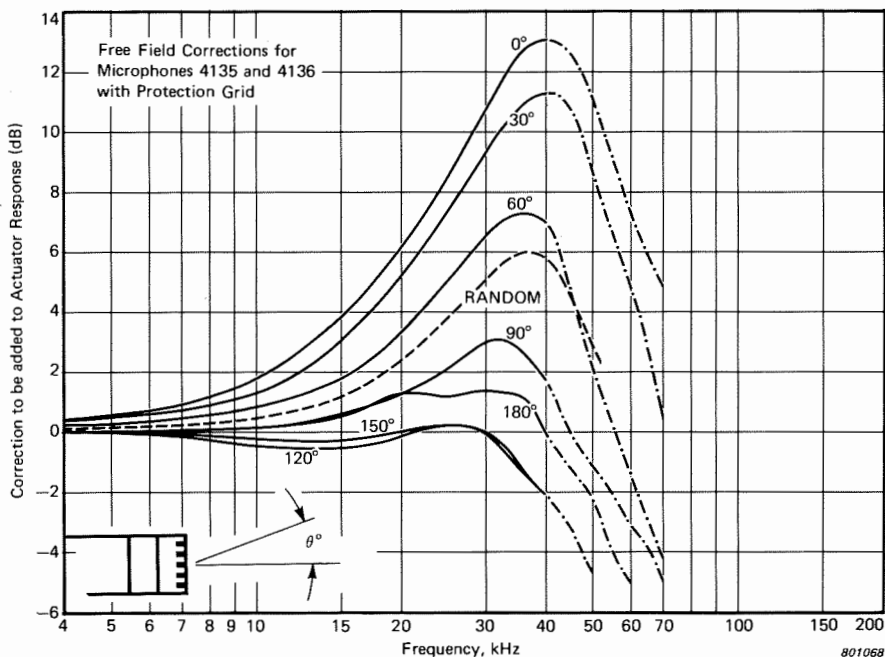


Fig.6.12. Free-field correction curves for quarter-inch Condenser Microphones Types 4135 and 4136 fitted with normal protection grid

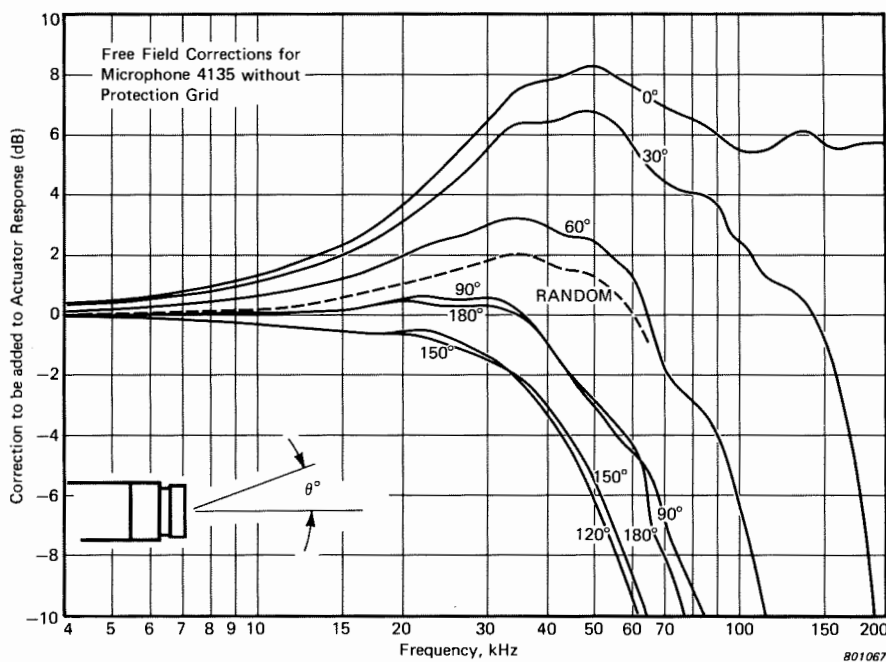


Fig.6.13. Free-field correction curves for quarter-inch Condenser Microphone Type 4135 without normal protection grid

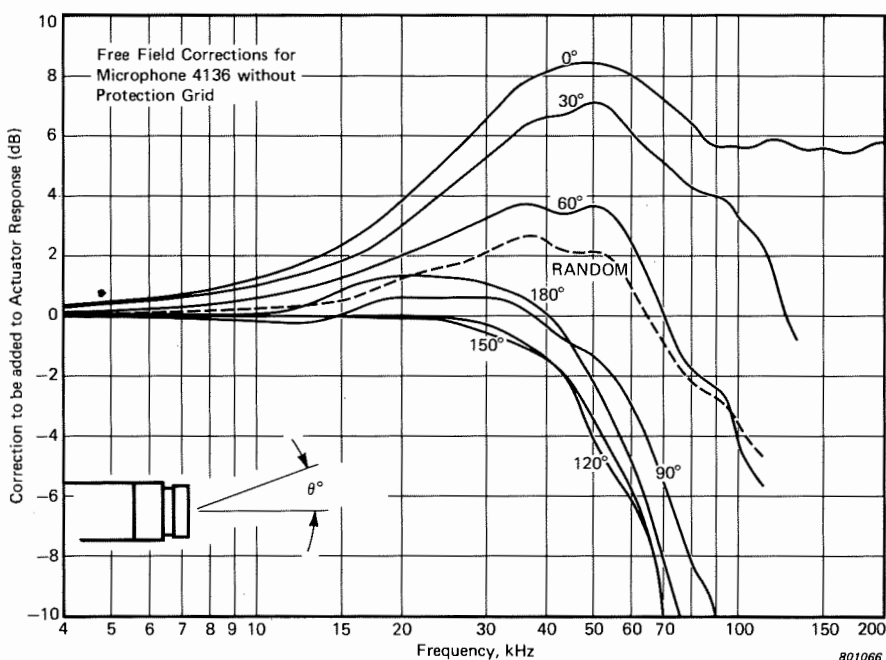


Fig.6.14. Free-field correction curves for quarter-inch Condenser Microphone Type 4136 without normal protection grid

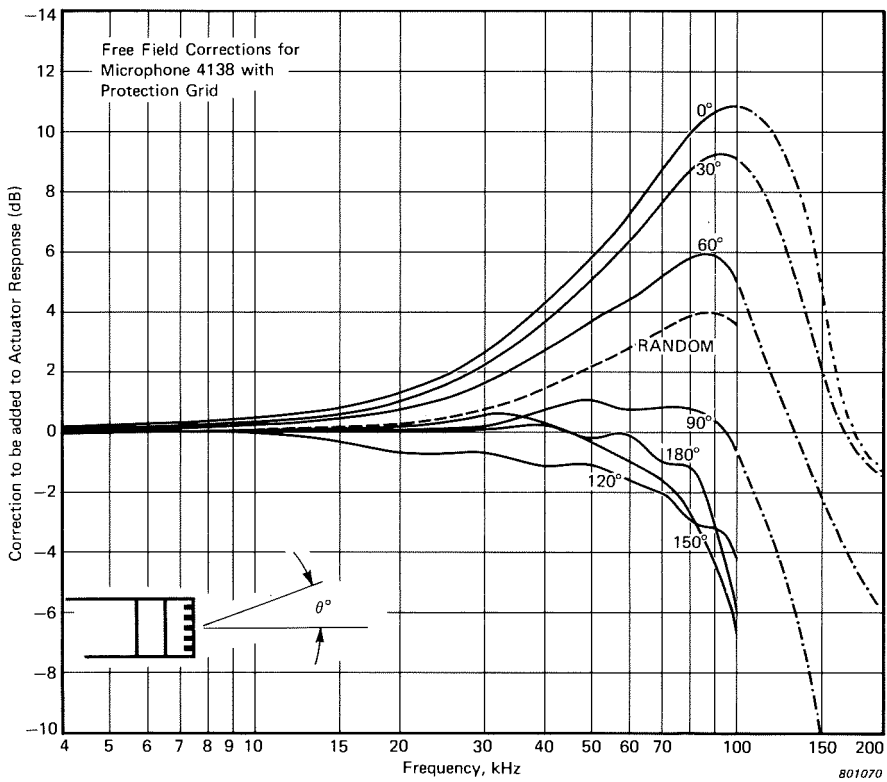


Fig.6.15. Free-field correction curves for eighth-inch Condenser Microphone Type 4138 fitted with normal protection grid

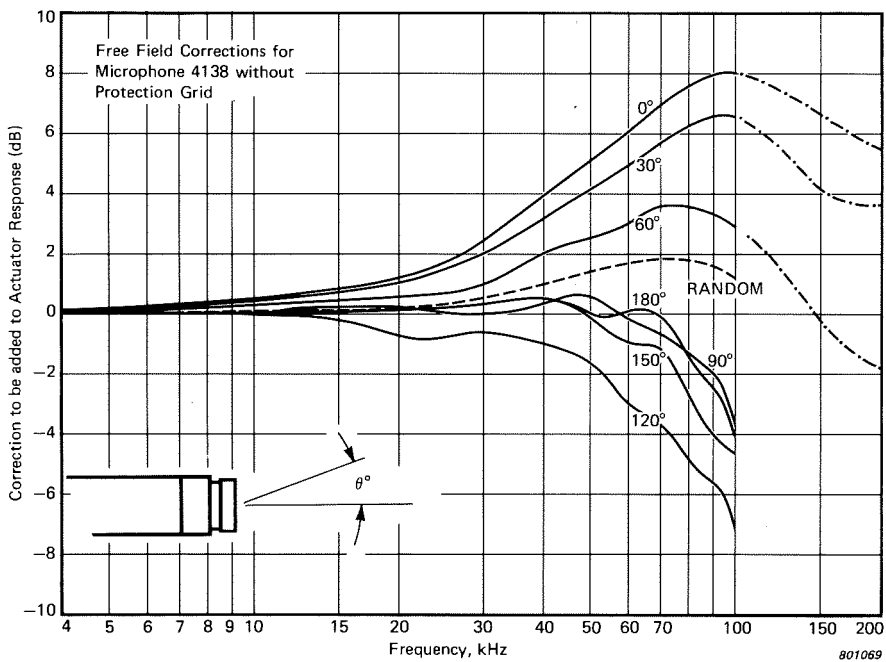


Fig.6.16. Free-field correction curves for eighth-inch Condenser Microphone Type 4138 without normal protection grid